For the presentation, everyone should know the basic LoRaWAN architecture and terminology, even if only one person worked on the ML part. Faculty can easily ask a basic question like “What is the difference between LoRa and LoRaWAN?” before moving to your project.

Here is a beginner-friendly LoRaWAN crash course focused on what is actually relevant to your project.

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1. What is IoT?

IoT = Internet of Things.

It means physical devices that can:

- sense something
- collect data
- communicate it over a network

Examples:

- temperature sensor
- smart electricity meter
- soil-moisture sensor
- GPS tracker
- air-quality sensor

Your project deals with IoT devices communicating using LoRaWAN.

---

2. What is LoRa?

LoRa = Long Range.

It is a wireless communication technology designed for:

- long distance
- low power
- small amounts of data

For example, a battery-powered temperature sensor might need to send only:

```
Temperature = 27°C
```

every few minutes.

It doesn't need Wi-Fi-level bandwidth.

Easy comparison

Technology	Range	Power usage	Typical use
Wi-Fi	Short	Higher	Internet, computers
Bluetooth	Very short	Low	Earbuds, peripherals
Cellular	Long	Higher	Phones
LoRa	Long	Very low	IoT sensors

---

3. What is LoRaWAN?

This is one of the most important questions.

LoRaWAN = Low Power Wide Area Network protocol.

LoRa is mainly the radio/physical communication technology.

LoRaWAN defines how the devices communicate as a network.

Easy way to remember:

> LoRa = how the signal is transmitted.
> LoRaWAN = how the IoT network operates.

So don't say:

> "LoRa and LoRaWAN are the same thing."

They aren't.

---

4. Why is LoRaWAN useful?

LoRaWAN is designed for IoT applications where devices:

- use little energy
- send small amounts of data
- may be far away from the gateway
- don't need continuous high-speed communication

Example:

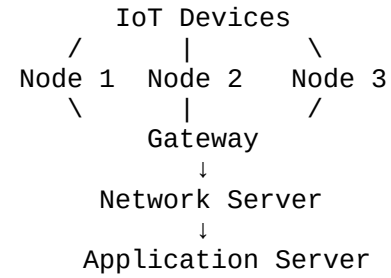
```
Temperature Sensor
        ↓
   small packet
        ↓
     Gateway
        ↓
 LoRaWAN Network
        ↓
```

```
Application
```

---

5. Basic LoRaWAN architecture

This is probably the most important diagram for your team to understand.



Let's understand each component.

---

6. End Device / Node

These are the actual IoT devices.

Examples:

```
Temperature sensor
Humidity sensor
Smart meter
GPS tracker
```

In your project, these are the nodes whose traffic you are monitoring.

You might have:

```
Node 1
Node 2
Node 3
Node 4
Node 5
Rogue Node
```

---

7. Gateway

The gateway receives LoRa radio messages from devices.

Think of it as a bridge:

```
LoRa wireless network
        ↓
     Gateway
        ↓
```

```
IP/Internet network
```

Important:

> A gateway usually does not decide whether a node is malicious.

It primarily receives and forwards traffic.

---

8. Network Server

The Network Server manages the LoRaWAN network.

It handles things such as:

- device/network management
- message processing
- security-related functions
- duplicate packet handling
- communication management

For your project, network traffic/data can be observed and analysed to identify unusual behaviour.

---

9. Application Server

The application server deals with the actual application data.

For example:

```
Sensor sends:

Temperature = 28°C
Humidity = 65%

          ↓

Application Server

          ↓

Dashboard
```

---

10. Complete flow

You should be able to explain this without looking at your PPT:

```
Sensor / Node
```

```
     ↓
LoRa radio
     ↓
Gateway
     ↓
Network Server
     ↓
Application Server
     ↓
User/Application
```

And your project sits around the traffic/data analysis part:

```
                LoRaWAN Network
                      ↓
                 Network Traffic
                      ↓
               Feature Extraction
                      ↓
                ML Detection
                      ↓
            Anomaly / Normal
                      ↓
             Suspicious Node
```

---

11. What is a LoRaWAN packet?

A packet is basically a unit of data transmitted over the network.

For example, a sensor might send:

```
Device → Gateway

Temperature = 28°C
Humidity = 65%
```

The actual LoRaWAN frame contains additional protocol information.

Your dataset contains information about this traffic that can be used as features for ML.

---

12. What is telemetry?

This may come up during your presentation.

Telemetry = measurements/data collected from a device and transmitted to another system.

Examples:

```
Temperature
Humidity
Battery level
Signal strength
Packet timing
```

In your project, when someone says:

> "What telemetry are you using?"

You can answer:

> "We use network and communication-related telemetry such as RSSI, SNR, spreading factor, timing/inter-arrival information and frame-counter behaviour, depending on the dataset."

Important distinction:

Not everything is necessarily sensor telemetry.

Your project is primarily analysing network traffic telemetry.

---

13. RSSI

RSSI = Received Signal Strength Indicator.

It tells us roughly how strong the received radio signal is.

Usually expressed in dBm.

Example:

```
-50 dBm → stronger signal
-100 dBm → weaker signal
```

Don't say:

> "-100 is stronger because 100 is bigger."

The values are negative.

The value closer to zero generally represents a stronger received signal.

Why useful?

A sudden unusual change in signal characteristics may be useful as one feature for anomaly detection.

But:

> RSSI alone cannot identify a rogue node.

---

14. SNR

SNR = Signal-to-Noise Ratio.

It indicates how strong the useful signal is compared with background noise.

Higher SNR generally means a cleaner signal.

Example:

```
Good signal → easier to distinguish from noise
Poor signal → more difficult
```

Again, SNR alone doesn't prove that a node is malicious.

---

15. Spreading Factor

Usually written as:

SF7, SF8, SF9 ... SF12

Spreading Factor is a LoRa modulation parameter.

In general:

Lower SF

```
Faster transmission
Less airtime
Usually shorter range
```

Higher SF

```
Slower transmission
More airtime
Usually greater range
```

So:

> Higher spreading factor generally gives longer range but lower data rate.

---

16. Airtime

Airtime = how long a packet occupies the radio channel.

Higher spreading factors generally result in longer airtime.

Why does this matter?

Because if a rogue device sends huge numbers of packets, it can consume a significant amount of the available communication time.

That can contribute to abnormal network behaviour.

---

17. Frame Counter — FCnt

This one is particularly relevant to your project.

FCnt = Frame Counter.

It helps keep track of frames sent by a device.

Think:

```
Packet 1 → FCnt 1
Packet 2 → FCnt 2
Packet 3 → FCnt 3
Packet 4 → FCnt 4
```

If the sequence suddenly behaves strangely, it can be an indicator worth investigating.

However:

> FCnt should not simply be treated as "high FCnt = attack."

And that's related to the FCnt leakage issue in your current synthetic dataset.

---

18. Uplink vs Downlink

Another very common question.

Uplink

```
Device → Gateway
```

The device sends data toward the network.

Example:

```
Temperature sensor → Gateway
```

Downlink

```
Gateway → Device
```

The network sends information back to the device.

Easy memory trick:

> Up = device to network
> Down = network to device

---

19. What is a rogue node?

Now connect LoRaWAN to your project.

A rogue node is a device that participates in the network but behaves in a way that is suspicious, abnormal, unauthorized, or malicious.

For example, it might:

- transmit excessively
- generate unusual traffic
- manipulate communication behaviour
- produce abnormal frame patterns

Your project asks:

> Can machine learning identify unusual traffic and help identify the node responsible?

---

20. What is an attack?

An attack is a deliberate attempt to disrupt, exploit, or misuse the network.

Examples relevant to LoRaWAN research include:

Replay attack

Previously captured messages are transmitted again.

```
Original message
       ↓
Captured
       ↓
Sent again
```

Flooding

An attacker generates excessive traffic.

```
Normal:
Node → 1 packet

Attack:
Node → packet
      packet
      packet
      packet
      packet...
```

Join-request flooding

An attacker sends a very large number of join requests.

This can put unnecessary load on the network.

---

21. Rogue node vs attack

These are not exactly the same thing.

Attack

The malicious behaviour.

Rogue node

The device performing the suspicious behaviour.

For your project:

```
Attack behaviour
      ↓
Abnormal traffic
      ↓
ML detects anomaly
      ↓
Identify node
      ↓
Rogue node
```

This distinction is worth remembering.

---

22. What is an anomaly?

An anomaly is simply:

> Something that significantly differs from expected/normal behaviour.

Example:

Normal node:

```
1 packet / 30 seconds
```

Suddenly:

```
100 packets / second
```

That's potentially anomalous.

But:

> Anomaly ≠ automatically attack.

This is a very good thing to say if faculty challenges you.

A legitimate device can also behave unusually because of:

- interference
- poor signal
- network conditions
- device malfunction

---

23. Why machine learning?

A simple traditional approach might be:

```
IF packets > 100
THEN attack
```

But real networks are more complicated.

Instead, ML can consider several characteristics:

```
RSSI
SNR
SF
Timing
FCnt behaviour
       ↓
     ML model
       ↓
Normal / Anomaly
```

So you're looking at patterns across multiple features.

---

24. Supervised vs unsupervised

Your team should absolutely know this.

Supervised learning

Training data has labels.

```
Features → Label
```

```
RSSI, SNR, SF → Normal
RSSI, SNR, SF → Rogue
```

Your:

- Random Forest
- XGBoost

are supervised approaches.

Unsupervised anomaly detection

The algorithm isn't simply given every attack label.

It tries to identify unusual behaviour.

Your:

Isolation Forest

is used as an anomaly-detection approach.

---

25. Why LoRaWAN is challenging for security

LoRaWAN is attractive for IoT because devices are:

- low power
- widely distributed
- often unattended
- communicating wirelessly

This creates security challenges.

A compromised device could potentially generate abnormal traffic.

And because there can be many devices, manually monitoring everything is difficult.

That's where automated anomaly detection can help.

---

26. What does your project actually contribute?

Don't say:

> "We created a completely new LoRaWAN security protocol."

You didn't.

Your contribution is more like:

```
LoRaWAN traffic
      ↓
Feature extraction
      ↓
Machine-learning models
      ↓
Anomaly detection
      ↓
Suspicious-node identification
```

```
      ↓
Visualization/dashboard
```

You're evaluating whether ML can help detect abnormal LoRaWAN behaviour.

---

27. The 10 things EVERY team member should memorize

If you have limited time tonight, make everyone learn these:

1. **LoRa = wireless technology**
2. **LoRaWAN = networking protocol**
3. **End device = IoT sensor/node**
4. **Gateway = receives LoRa messages and forwards them**
5. **Network Server = manages/processes the network**
6. **Uplink = device → network**
7. **Downlink = network → device**
8. **RSSI = received signal strength**
9. **SNR = signal quality relative to noise**
10. **Anomaly = behaviour that differs significantly from normal**

Then learn:

```
FCnt = Frame Counter
SF = Spreading Factor
IAT = Inter-arrival time
```

---

28. The 30-second LoRaWAN explanation

If faculty suddenly says:

> "Explain LoRaWAN."

Say:

> "LoRaWAN is a low-power wide-area networking protocol designed for IoT devices. End devices such as sensors communicate wirelessly using LoRa with gateways. The gateways forward the data to the network server, which manages and processes the network communication, and the application server handles the application data. LoRaWAN is useful for applications where devices need long-range communication, low power consumption and relatively small amounts of data."

That's a very safe answer for a beginner-level presentation.

---

29. Then connect it to YOUR project

If they ask:

> "Okay, where does your project fit into this?"

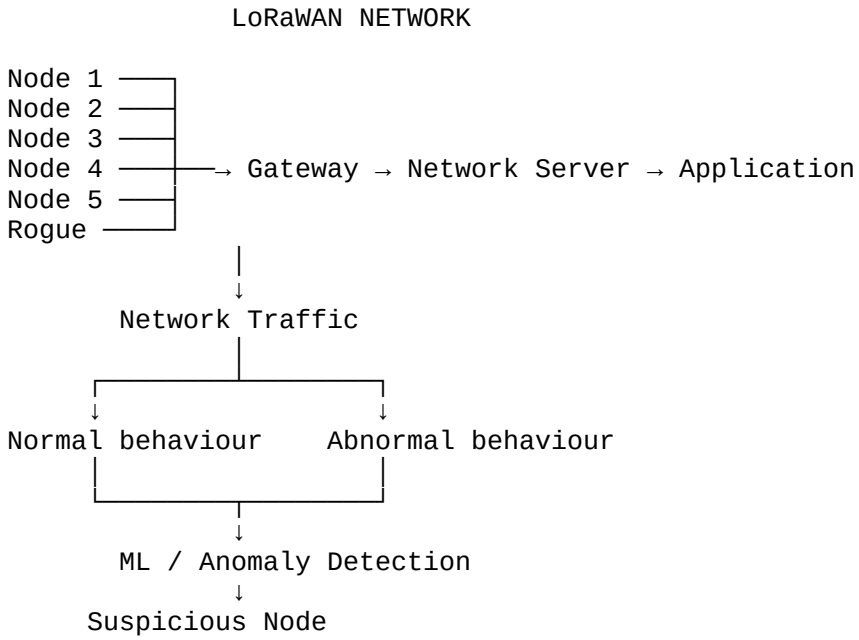
Say:

> "Our project analyses the traffic generated by these LoRaWAN nodes. We extract network-related features such as RSSI, SNR, spreading factor, timing and frame-counter behaviour, and use machine-learning models to identify traffic that differs from normal behaviour. We then want to use those anomaly results to identify the node that may be behaving like a rogue node."

That connects the LoRaWAN basics → problem → ML solution perfectly.

One final mental picture

Everyone on your team should have this in their head:



That is essentially your entire project story in one diagram.